

LONGEVITY OS

Evidence-Based Healthspan Blueprint

Busy Urban Adult • Singapore Edition • 2026

LONGEVITY = longer life + longer healthspan

Three outcomes, driven by one practical intervention layer

1 PREVENT DISEASE

Cardiovascular • cancer • metabolic
brain • kidney • infection • frailty

2 SLOW BIOLOGICAL AGING

Hallmarks of Aging • mitochondria
autophagy • senescence • inflammation

3 BUILD & PRESERVE RESERVE

VO₂max • muscle • strength • bone
balance • mobility • cognition

4 OPTIMIZE FUNDAMENTALS

Exercise • food • sleep • body composition
no tobacco • prevention • environment

The central idea

Maximize healthy functional years by preventing major disease, slowing damaging aging processes, preserving physical and cognitive reserve, and using the simplest high-ROI fundamentals that can be sustained for decades.

Evidence synthesized through 22 August 2026

Educational resource — not individualized medical advice

SECTION 00

How to use this report

A concise synthesis of the entire research thread — deduplicated, evidence-ranked, and localized for real life.

Important scientific boundary

Current medicine can meaningfully reduce premature disease, improve fitness and metabolic health, preserve function, and modify some biological processes associated with aging. It has not demonstrated that humans can stop aging, reverse whole-organism age, or become immortal. Radical rejuvenation remains a frontier research goal.

This report deliberately separates three things that longevity content often mixes together:

- Outcomes — living longer, remaining healthy and independent, avoiding major disease.
- Biology — hallmarks of aging, mitochondria, autophagy, senescence, inflammation and related mechanisms.
- Interventions — exercise, food, sleep, risk-factor control, screening, supplements and experimental therapies.

Decision rule

A recommendation moves upward only when it combines strong human evidence, meaningful expected benefit, low risk, reasonable cost, low time burden and long-term adherence. Mechanistic excitement alone is never allowed to outrank human outcomes.

Contents

#	Section
01	Executive summary — The Longevity 10
02	The four-pillar master model
03	Evidence hierarchy + Real-World Fit
04	Pillar 1 — Prevent disease
05	Pillar 2 — Slow biological aging
06	Mitochondrial health
07	Pillar 3 — Preserve physiological reserve
08	Pillar 4 — Fundamentals
09	Exercise protocol for busy adults
10	Nutrition + Singapore hawker strategy
11	Sleep, social health and recovery
12	Singapore environment — heat, UV and haze
13	Biomarkers, screening and prevention
14	Supplements — core to skip
15	Frontier geroscience
16	The 80/20 protocol
17	Evidence notes + references

SECTION 01

Executive summary

If someone reads only three pages, these are the pages.

The research converges on a simple conclusion:

Longevity is mostly not a shopping problem.

The highest-return actions are ordinary: do not smoke, move and train, maintain healthy adiposity and metabolic health, control blood pressure and atherogenic lipoprotein exposure, eat a high-quality diet, sleep adequately, and use evidence-based screening and vaccination. Expensive supplements and experimental geroprotectors come much later.

The Longevity 10

1. Never smoke; if you smoke, quitting outranks almost every optimization.
2. Move every day. Around 7,000 daily steps is a useful practical target — not a magic threshold. [3]
3. Accumulate 150–300 minutes of moderate aerobic activity per week, or vigorous equivalent. [2]
4. Strength-train all major muscle groups at least twice per week. [2,4]
5. Maintain a healthy waist and body-fat trajectory while preserving muscle.
6. Eat mostly minimally processed, fiber-rich foods with adequate protein.
7. Protect regular, restorative sleep — roughly 7–9 hours for most adults. [10]
8. Know and appropriately manage blood pressure, lipids/ApoB-related risk and glucose.
9. Minimize alcohol, excess sodium, harmful UV and avoid unnecessary carcinogen exposure.
10. Use evidence-based vaccination and age/risk-appropriate screening.

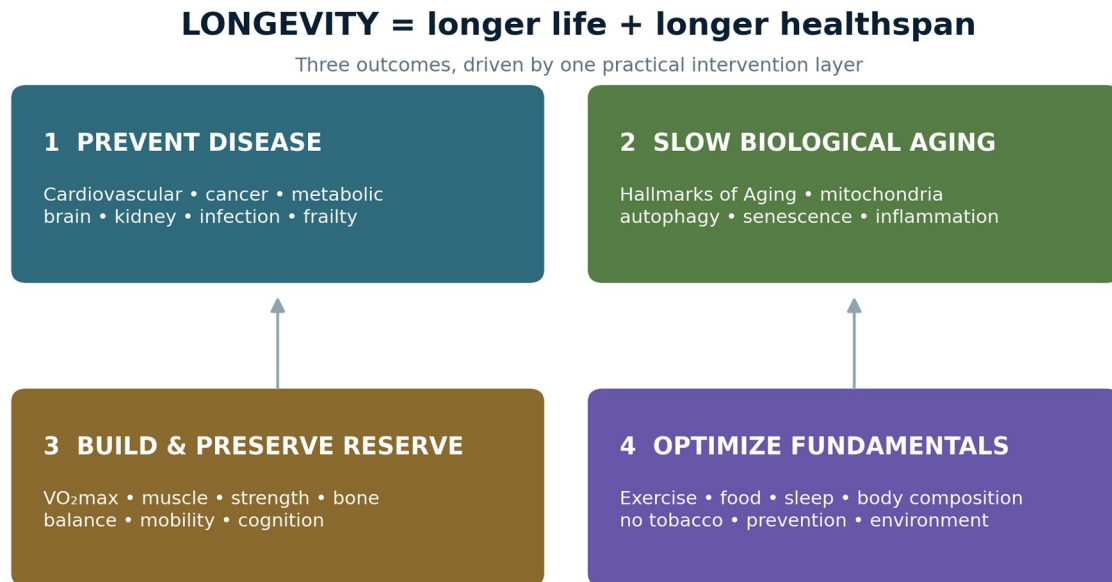
Highest-ROI action ranking

Tier	Action	Impact	Cost	Time
CORE	Do not smoke / quit	Very high	Saves money	Very low
CORE	Blood-pressure control	Very high	\$	Very low
CORE	Aerobic activity / fitness	Very high	\$0	Moderate
CORE	Resistance training	Very high	\$0–\$	Low
CORE	Daily movement / ~7k+ steps	High	\$0	Low
CORE	Healthy waist / visceral fat	Very high	\$0	Low
CORE	LDL/ApoB-related risk control	Very high	\$–\$\$	Very low
CORE	High-quality dietary pattern	Very high	\$	Low
CORE	Healthy glucose metabolism	Very high	\$	Low
CORE	Adequate regular sleep	High	\$0	Low
CORE	Low / no alcohol for cancer prevention	High	\$0	Very low
CORE	Vaccination + appropriate screening	Very high	\$–\$\$	Very low

SECTION 02

The four-pillar master model

One hierarchy that contains longevity, anti-cancer prevention, mitochondrial health, “reverse aging,” and functional health without duplication.



How the pillars differ

Pillars 1–3 are outcomes. Pillar 4 is the practical intervention layer that drives all three. Mitochondrial health is primarily a mechanism inside aging biology and reserve; cancer prevention is primarily a disease-prevention branch; “reverse aging” is not a separate pillar.

Pillar	Simple meaning	Includes
1. Prevent Disease	Reduce premature death and disability	CVD, cancer, diabetes, fatty liver, dementia, CKD, infection, frailty
2. Slow Biological Aging	Maintain cellular repair and resilience	12 Hallmarks of Aging, mitochondria, autophagy, proteostasis, senescence, inflammation
3. Build & Preserve Reserve	Keep capacity to withstand aging, illness and inactivity	VO ₂ max, muscle, strength, power, bone, balance, mobility, cognition
4. Optimize Fundamentals	Control the everyday inputs	Exercise, diet, sleep, body composition, tobacco/alcohol, BP/lipids/glucose, prevention, environment

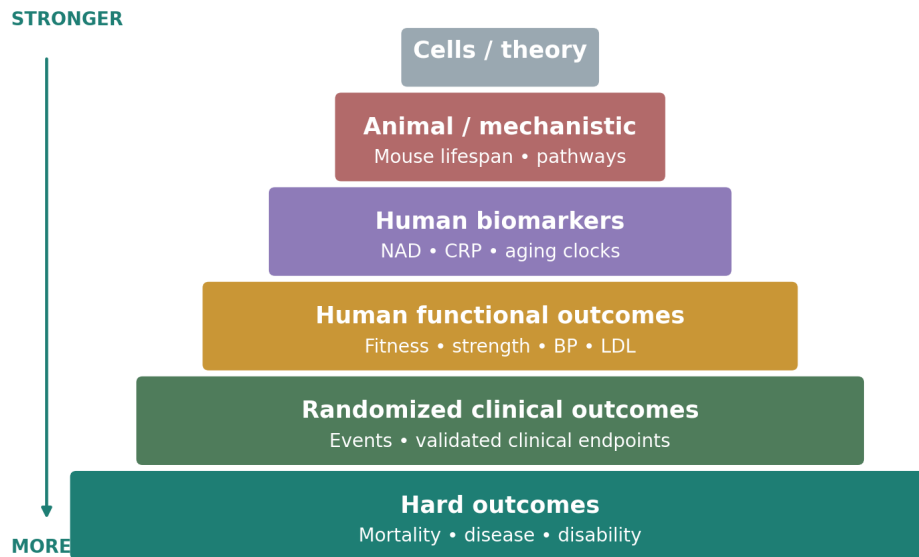
One-sentence model

Maintain metabolic and cellular resilience while minimizing accumulated damage — and preserve enough physical and cognitive reserve to survive the stresses of aging.

SECTION 03

Evidence hierarchy + Real-World Fit

Human outcomes outrank pathway stories. Sustainability outranks theoretical perfection.



Evidence labels used in this report

Label	Meaning
CORE	Strong human evidence + high practical value. Prioritize first.
CONDITIONAL	Good value for specific people, deficiencies or risk profiles.
OPTIONAL	Reasonable incremental benefit, but not foundational.
EXPERIMENTAL	Interesting human/preclinical science without proven longevity outcomes.
SKIP	Weak evidence, poor economics, unnecessary complexity or unfavorable trade-off.

Real-World Fit filter

Every intervention should be judged a second time after the science:

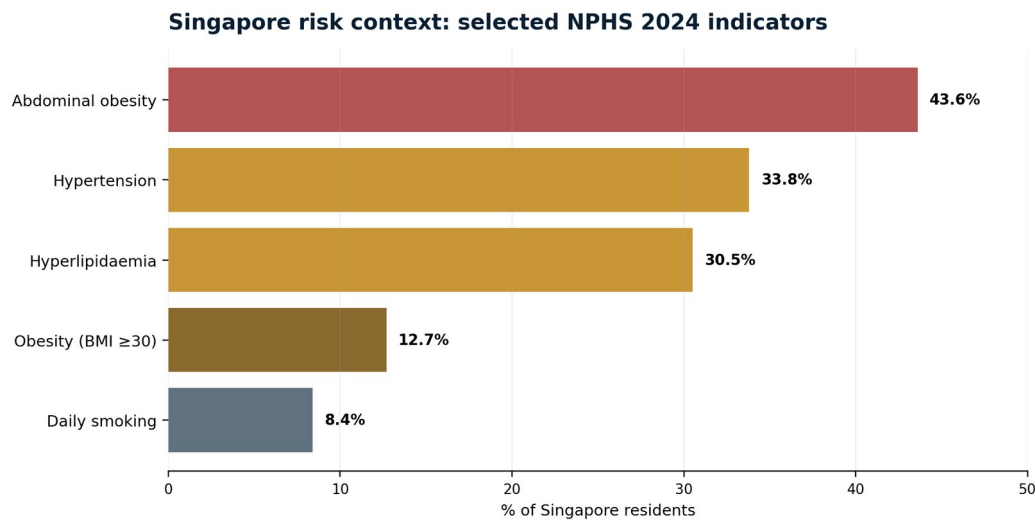
Dimension	Question
Evidence	Does it improve meaningful human outcomes?
Impact	How much could it matter?
Safety	What are the downside and contraindications?
Cost	What does it cost locally?
Time	How many hours per week?
Availability	Can normal people access it?
Environment	Does heat, UV, haze or infrastructure change implementation?
Adherence	Can it realistically be sustained for decades?

SECTION 04

Pillar 1 — Prevent major disease

The fastest way to improve expected longevity is to reduce the risks that actually cause disease and death.

Singapore makes this especially concrete. In 2024, cancer accounted for 26.5% of deaths, pneumonia 23.0%, ischaemic heart disease 19.6%, stroke 5.6% and hypertensive diseases 3.2%. [26]



NPHS 2024 also reported hypertension in 33.8%, hyperlipidaemia in 30.5%, obesity in 12.7%, and abdominal obesity in 43.6% of residents aged 18–74. [18]

Cardiovascular disease

- Blood pressure: randomized evidence shows meaningful reductions in cardiovascular events when elevated blood pressure is effectively lowered. [5]
- Atherogenic lipoproteins: LDL-containing ApoB particles are causal drivers of atherosclerotic cardiovascular disease. [6]
- Lp(a): the 2026 ACC/AHA dyslipidaemia guideline recommends measuring Lp(a) at least once in adulthood; selective ApoB measurement can refine risk. [28]
- Exercise, healthy body composition, diet, sleep and tobacco avoidance support risk reduction, but medication should not be rejected when clinically indicated.

Metabolic disease

- Focus on insulin sensitivity, healthy glucose regulation, visceral/ectopic fat and preservation of muscle — not just scale weight.
- High-value measures are waist, weight trajectory, blood pressure, fasting glucose/HbA1c and lipids; expensive body-composition testing is optional.

Cancer prevention

- Avoid tobacco and second-hand smoke.
- Maintain healthy body composition and stay physically active.
- Favor a plant-rich, fiber-rich dietary pattern; limit processed meat and avoid making alcohol a “health food.” [7]
- Protect against excessive UV and relevant occupational/environmental carcinogens.
- Use vaccination against oncogenic infections when indicated (e.g., HPV, hepatitis B) and follow evidence-based screening. WCRF explicitly advises against using supplements for cancer prevention. [7]

SECTION 05

Pillar 2 — Slow biological aging

Use the Hallmarks of Aging as a map of biology — not as twelve separate consumer protocols.

The 2023 Hallmarks of Aging update describes 12 interconnected hallmarks. [1]

Hallmark	Meaning	Practical stance today
Genomic instability	DNA damage and repair failure	Avoid carcinogens; fundamentals; no proven “DNA repair supplement”
Telomere attrition	Loss of protective chromosome ends	Do not chase telomere supplements; focus on overall health
Epigenetic alterations	Changes in gene regulation	Exercise/metabolic health; aging clocks remain research tools
Loss of proteostasis	Poor protein quality control	Exercise; adequate nutrition
Disabled macroautophagy	Impaired cellular recycling	Exercise; healthy energy balance
Deregulated nutrient sensing	mTOR/AMPK/insulin/IGF-1 regulation	Healthy weight, activity; avoid chronic overnutrition
Mitochondrial dysfunction	Energy + quality-control decline	Aerobic + resistance exercise
Cellular senescence	Damaged cells persist and signal	Exercise; senolytics experimental
Stem-cell exhaustion	Reduced regenerative capacity	Maintain health/function; therapies experimental
Altered intercellular communication	Disrupted systemic signaling	Exercise/metabolic health
Chronic inflammation	Persistent low-grade inflammatory signaling	Healthy adiposity, exercise, diet, no smoking
Dysbiosis	Microbiome/ecosystem disruption	Fiber/diverse foods; condition-specific treatment

What “reverse aging” should mean today

Type	Status	Interpretation
Disease-risk reversal	Very achievable	BP, lipids, glucose, visceral fat, fatty liver risk
Functional rejuvenation	Very achievable	Become fitter, stronger, leaner and more capable than years earlier
Biomarker rejuvenation	Sometimes possible	Some aging clocks and molecular markers can move
Whole-organism age reversal	Not established	No therapy has proven durable systemic age reversal in humans

2026 biomarker update

A Nature Medicine analysis harmonized 51 human intervention studies and found that mortality/pace-of-aging clocks such as DunedinPACE and GrimAge-family measures were more consistently responsive than older chronological-age clocks. This improves trial design — but does not prove that changing a clock extends lifespan. [17]

SECTION 06

Mitochondrial health

Mitochondria matter — but the best “mitochondrial stack” is still mostly exercise and metabolic health.

Healthy mitochondrial biology requires four linked capabilities:

Function	What it means
Make	Biogenesis — creating new mitochondria
Use	Efficient ATP production and metabolic flexibility
Maintain	Fusion/fission, redox signaling and quality control
Remove	Mitophagy — clearing dysfunctional mitochondria

Highest-value interventions

- Aerobic training — the clearest practical intervention for mitochondrial content and oxidative capacity.
- Intervals / vigorous work — useful for improving cardiorespiratory ceiling when tolerated.
- Resistance training — supports muscle, glucose disposal and mitochondrial adaptations.
- Healthy energy balance and insulin sensitivity — avoid chronic metabolic overload.
- Adequate sleep and nutrition — support recovery; neither requires a special mitochondrial product.

Mitochondrial supplement reality check

Intervention	Verdict	Bottom line
Creatine	CONDITIONAL	Strong performance/lean-mass adjunct with resistance training; not proven to extend lifespan. [12]
CoQ10	CONDITIONAL	Useful in selected clinical contexts; weak case as universal longevity supplement.
Magnesium	CONDITIONAL	Correct deficiency / selected indications; not a mitochondrial anti-aging drug.
Urolithin A	OPTIONAL / EXPERIMENTAL	Interesting mitophagy/function signals; hard longevity outcomes absent.
NR / NMN	EXPERIMENTAL	Can raise NAD; anti-aging clinical effectiveness remains unproven.
PQQ / antioxidant stacks	SKIP / LOW PRIORITY	Mechanistic appeal without comparable human outcome evidence.

Rule of thumb

“Raises NAD,” “boosts autophagy,” or “improves a mitochondrial biomarker” is not equivalent to reducing disease, disability or mortality.

SECTION 07

Pillar 3 — Build and preserve physiological reserve

Longevity is not only avoiding disease. It is arriving at older age with enough capacity to survive illness, surgery, falls and periods of inactivity.

Reserve	Why it matters	Practical lever
Cardiorespiratory fitness	VO ₂ max / aerobic capacity	Build with regular aerobic work + some vigorous training
Muscle mass	Metabolic and functional reserve	Resistance training + adequate protein
Strength	Ability to perform work and maintain independence	Progressive resistance training
Power	Fast force production; increasingly relevant with age	Safe explosive intent / faster concentric work when appropriate
Bone	Fracture resistance	Loading, resistance exercise, adequate nutrition
Balance / mobility	Fall prevention and independence	Practice progressively, especially with age
Cognition	Mental reserve	Exercise, vascular health, sleep, hearing, learning/social engagement

Protein: useful, not magical

Protein supports muscle maintenance and training adaptation. In resistance-training studies, benefit from additional protein appears to plateau around total intakes near 1.6 g/kg/day; this is an approximate ceiling for maximizing training-related lean-mass gains, not a universal minimum requirement. [11]

Practical interpretation: prioritize adequate daily protein, especially when training, dieting or aging. Food first; whey is primarily a convenience tool.

SECTION 08

Pillar 4 — Optimize the fundamentals

The everyday inputs that drive all three outcome pillars.

Fundamental	Why it matters
Exercise	Aerobic fitness, mitochondrial adaptation, BP/glucose control, muscle/bone, cancer risk
Nutrition	Energy balance, body composition, cardiometabolic health, fiber/micronutrients
Sleep	Recovery, cognition, cardiometabolic regulation, daytime function
Body composition	Visceral/ectopic fat vs muscle reserve
Tobacco / alcohol	Major preventable disease and cancer exposures
BP / lipids / glucose	Causal and actionable risk factors
Vaccination / screening	Prevent infection-related cancer and detect important disease early
Environment	UV, heat and air quality change implementation in Singapore
Social / mental health	Supports function and adherence; mortality evidence is mostly observational

Less is more

The purpose of optimization is not to add more interventions. It is to remove bottlenecks: inactivity, smoking, poor BP/lipid control, excess visceral fat, weak fitness, poor sleep, low-quality diet and missed prevention.

SECTION 09

Exercise protocol for a busy adult

Build the minimum effective system first; only add volume when it improves results without harming adherence.

Evidence anchor

WHO recommends 150–300 minutes/week of moderate aerobic activity (or 75–150 vigorous) and muscle-strengthening work on at least two days/week. [2] A 2025 dose-response meta-analysis found substantial health associations around 7,000 daily steps versus 2,000, while benefits began well below 7,000 and often continued beyond it. [3]

Level	Who	Prescription
MINIMUM	For a sedentary / overloaded week	20–30 min walking most days + 2 short full-body strength sessions
HIGH ROI	Best default	~7k–10k steps/day when practical + 2–3 strength sessions + 150–300 min aerobic/week
ENTHUSIAST	Only after the basics are easy	Add structured VO ₂ max/interval work, more aerobic volume, power/mobility — without sacrificing recovery

Sample busy-person week

Day	Plan
Mon	35 min full-body strength
Tue	30–40 min brisk cardio / Zone-2-like effort
Wed	Normal movement + walking; optional 5-min exercise snacks
Thu	35 min full-body strength
Fri	30–40 min cardio
Sat	45–60 min enjoyable movement / sport / walk
Sun	Recovery + normal movement

Full-body movement template

One squat/leg press • one hinge • horizontal push • row • vertical pull • overhead press • carry/core. Progress gradually. A gym is optional; bodyweight, bands and dumbbells can cover the basics.

When time collapses

Short “exercise snacks” — stairs, brisk walking, squats, push-ups or similar bouts — are a useful fallback for inactive people. They should complement, not permanently replace, progressive strength and aerobic training.

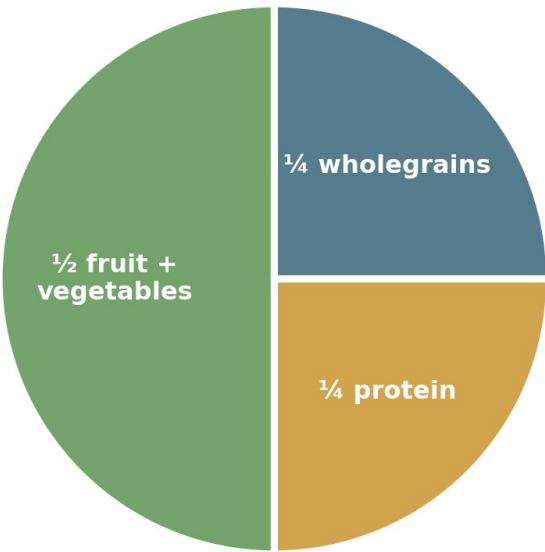
SECTION 10

Nutrition + Singapore hawker strategy

Use ordinary food, not “longevity food.” Make the default meal good enough to repeat for decades.

Simple meal rule

Simple plate rule for most meals



Singapore's My Healthy Plate uses the same Quarter–Quarter–Half logic: 1/4 wholegrains, 1/4 protein and 1/2 fruit/vegetables. [20]

Dietary pattern

- Base most meals on vegetables/fruit, whole grains, nuts/seeds, fish and other minimally processed protein sources, and unsaturated fats.
- Get adequate protein and fiber; use whole fruit more often than juice.
- Limit processed meat, frequent sugar-sweetened drinks and chronically energy-dense ultra-processed diets.
- Mediterranean-style patterns have among the strongest observational and trial-supported longevity/cardiovascular evidence, but the foods can be localized rather than imported literally. [29]

The Singapore hawker algorithm

Step	Default
1. Protein	Fish, chicken, eggs, tofu, lean meat; fewer processed/fried defaults
2. Vegetables	Ask for extra vegetables or pair the meal with greens/fruit
3. Carbohydrate	Wholegrain when easy; otherwise use a sensible portion of normal rice/noodles
4. Sauce / soup	Less gravy and sauce; do not automatically finish salty broth
5. Drink	Water or low/no-sugar drink as default

Singapore’s sodium bottleneck

The 2022 National Nutrition Survey found average sodium intake around 3,620 mg/day and about 90% of residents exceeding the 2,000 mg/day recommendation. Major sources included salt, sauces, seasonings, soups, gravy-based dishes, flavored rice and noodles. [19] WHO recommends <2,000 mg/day sodium for adults. [8]

Good hawker templates

Meal	How to improve it
Mixed vegetable rice / cai fan	2 vegetables + 1 lean protein + rice; minimal gravy
Sliced fish soup	Extra vegetables; do not finish all broth
Yong tau foo	Mostly vegetables/tofu/egg/non-fried items; limit soup
Chicken rice	Extra greens; moderate rice; less sauce; skin optional
Chapati-based meal	Whole-wheat chapati + vegetables + lean protein

Budget longevity pantry

Oats • eggs • tofu/soy • frozen vegetables • whole fruit • rice/whole grains • canned fish • chicken • plain dairy • nuts/seeds. Convenience foods such as whey or frozen produce are worth paying for when they improve adherence.

The goal is not dietary purity. A sustainable 80–90% sensible default usually beats a perfect plan that collapses.

SECTION 11

Sleep, social health and recovery

Protect the ability to recover; do not sacrifice sleep to squeeze in more “optimization.”

For most adults, 7–9 hours is a reasonable sleep-duration target. [10] Duration is only one dimension; regularity, timing, quality and sleep disorders also matter.

Priority	Action
1	Protect enough sleep opportunity — schedule it like a non-negotiable input.
2	Keep wake time reasonably consistent; use morning light and daytime activity.
3	Make the room cool, dark and quiet — especially useful in Singapore’s heat/humidity.
4	Avoid using alcohol as a sleep tool; manage caffeine timing if it delays sleep.
5	If persistent loud snoring, gasping, witnessed apneas or daytime sleepiness occur, assess for sleep apnea rather than buying more supplements.

Social and mental health

Loneliness, isolation, depression and chronic stress are associated with worse health outcomes, although much of the mortality evidence is observational. Treat connection, purpose and mental-health care as part of healthspan — without pretending they have an exact “years added” dose.

Recovery principle

A longevity routine that chronically worsens sleep, stress or social life is not automatically a better longevity routine.

SECTION 12

Singapore environment — heat, UV and haze

The environment changes when and where you exercise, not whether movement matters.

Heat / humidity

WBGT	Level	Practical action
WBGT <31°C	Low heat stress	Normal activity
31–<33°C	Moderate	Reduce prolonged outdoor activity; regular shade/indoor breaks
≥33°C	High	Minimize prolonged outdoor activity; use more frequent/longer cooling breaks

Singapore's 2026 public heat-stress guidance uses these thresholds for prolonged outdoor activity. [23]

UV

UV is commonly very high to extreme around 11am–3pm on relatively cloudless days. NEA recommends shade/umbrella, protective eyewear/hat and at least SPF 30 when exposure is significant. [24]

Haze / air quality

Use current 1-hour PM2.5 for immediate decisions and the 24-hour PSI forecast for forward planning. Healthy people should reduce prolonged/strenuous outdoor exertion when PSI is 101–200, avoid it at 201–300, and minimize outdoor activity above 300. [25]

10-second outdoor decision

Before a long/hard outdoor session: check WBGT + UV + 1-hour PM2.5/PSI + rain/lightning. If conditions are poor, move the session indoors. The training adaptation still counts.

SECTION 13

Biomarkers, screening and prevention

Measure what changes decisions. Do not confuse more testing with better prevention.

Tier	Examples
Tier A – high actionability	Blood pressure; standard lipid panel; glucose/HbA1c; weight + waist; activity/fitness; smoking status
Tier B – situational	ApoB; kidney/urine albumin; liver tests; ferritin/iron; B12; vitamin D; thyroid – based on context
Tier C – research / optimization	Lab VO ₂ max; DEXA; CGM in healthy people; epigenetic aging clocks
Tier D – low value as routine	Repeated tumor markers in healthy people; giant micronutrient panels; “mitochondrial age”; indiscriminate whole-body scans

Lipids: one useful 2026 update

The 2026 ACC/AHA dyslipidaemia guideline recommends Lp(a) at least once in adulthood and selective ApoB measurement to refine risk in appropriate people. [28]

Healthier SG advantage

For eligible Singapore Citizens, Healthier SG Screening is \$5 or less per screening visit at CHAS GP clinics, with \$0 for Pioneer Generation, \$2 for Merdeka Generation / CHAS Blue or Orange, and \$5 for other eligible citizens / CHAS Green; enrolled Healthier SG citizens can receive special subsidies at their enrolled clinic. [22]

Testing question

If I know this number, will it change an action proven to improve a meaningful outcome? If not, the test probably belongs lower in the stack.

Screening and vaccination

Use country-, age-, sex- and risk-aware guidelines rather than universal mega-panels. Appropriate cancer screening, cardiovascular/metabolic screening and nationally recommended vaccination usually provide far more value than exotic longevity diagnostics.

SECTION 14

Supplements — core, conditional, optional, experimental, skip

Supplements should solve a real problem: deficiency, dietary gap, convenience, performance or a specific clinical indication.

Intervention	Verdict	Interpretation
Creatine monohydrate	CONDITIONAL	Good evidence as a resistance-training adjunct; modest additional lean-mass gain. Not a lifespan drug. [12]
Psyllium / supplemental fiber	CONDITIONAL	Useful when diet is fiber-poor and for selected lipid/glucose goals.
Vitamin D	CONDITIONAL	Correct deficiency / support indicated bone care; not universal anti-aging therapy.
Omega-3	CONDITIONAL	Context depends on fish intake, dose/formulation and cardiovascular risk.
Magnesium	CONDITIONAL	Correct deficiency / selected indications; not universal longevity treatment.
Protein powder / whey	CONDITIONAL	Convenience for meeting protein needs; food works equally well.
CoQ10	CONDITIONAL / OPTIONAL	Selected contexts; weak case as a universal longevity supplement.
Glycine / taurine	OPTIONAL / EXPERIMENTAL	Interesting biology; human lifespan evidence absent.
Curcumin / polyphenol extracts	OPTIONAL	May affect intermediate markers; hard-outcome longevity evidence weak.
Probiotics	CONDITIONAL	Condition-specific; no universal "longevity microbiome" capsule.
NR / NMN	EXPERIMENTAL	Raises NAD, but anti-aging clinical effectiveness remains unproven.
Urolithin A	EXPERIMENTAL / OPTIONAL	Promising mitochondrial/mitophagy signals; no human lifespan evidence.
Fisetin / quercetin as senolytics	EXPERIMENTAL	Human senolytic outcome evidence is immature.
Large antioxidant stacks	SKIP	Poor ROI versus exercise and a high-quality diet; high doses can have downsides.

Supplement rule

Correct deficiencies and use targeted tools. Do not let a supplement stack displace exercise, sleep, healthy food, BP/lipid control or screening.

SECTION 15

Optional optimization + frontier geroscience

Interesting science belongs in the report — but behind the fundamentals, with uncertainty clearly labeled.

Optional tools

Tool	Verdict	Bottom line
Time-restricted eating / fasting	OPTIONAL	Can help some people control calories and metabolic health; not required for longevity.
Moderate caloric restriction	OPTIONAL / CONTEXTUAL	CALERIE improved cardiometabolic measures and modestly altered DunedinPACE, but long-term lifespan extension in humans is unproven. [13,14]
Sauna	OPTIONAL	Promising observational cardiovascular associations; hard randomized outcome evidence weaker than exercise/BP treatment.
Cold exposure	LOW PRIORITY	Small heterogeneous trials; no compelling longevity-outcome evidence.
Wearables / CGM	OPTIONAL	Useful when they change behavior; data collection alone does not improve health.

Frontier therapies

Therapy	Status	Current interpretation
Rapamycin / rapalogs	EXPERIMENTAL	Human studies show some physiological effects, but healthy-human lifespan/disease-delay benefit is not established; risks and optimal long-term dosing remain uncertain. [15]
Metformin in healthy non-diabetics	EXPERIMENTAL	Excellent diabetes drug; insufficient evidence to prescribe broadly for longevity.
Senolytics	EXPERIMENTAL	Strong animal rationale; human translation still early.
NAD manipulation	EXPERIMENTAL	Biochemical target engagement ≠ proven healthspan extension.
Partial cellular reprogramming	FRONTIER	First participant dosed in a Phase 1 OSK-based optic-neuropathy trial in June 2026. Safety/efficacy unknown; this is a historic translational milestone, not a consumer therapy. [16]
Gene / regenerative rejuvenation	FRONTIER	Potentially transformative but currently research-stage.

Bridge-generation mindset

The defensible strategy is not to assume immortality is available today. It is to preserve cardiovascular, metabolic, muscular and cognitive reserve so you are more likely to remain healthy enough to benefit if genuinely rejuvenative therapies arrive later.

SECTION 16

The 80/20 protocol

What to do today, this week, this year — and what not to overcomplicate.

Today

- Move: accumulate walking; take stairs; break up long sitting.
- Eat: protein + vegetables/fruit + sensible carbohydrate + healthy fat; at hawker centres, use less gravy/soup and choose water.
- Sleep: protect enough time for ~7–9 hours and a consistent wake time.
- Avoid: tobacco; excess alcohol; unnecessary midday UV.
- If training outdoors: check WBGT, UV and air quality first.

This week

- 2 full-body strength sessions
- 150+ minutes of moderate aerobic activity or equivalent progression toward it
- Several days near or above ~7,000 steps if practical
- Buy simple repeatable foods rather than specialty longevity products
- Schedule missing preventive care rather than another supplement order

This year

- Know your blood pressure, lipid profile and glucose status
- Measure/monitor waist and body-weight trajectory
- Consider a once-in-adulthood Lp(a) test as part of cardiovascular risk assessment
- Stay current on appropriate vaccination and cancer screening
- Reassess fitness, strength, sleep quality and adherence — not only laboratory numbers

Do this instead

Instead of	Do	Why
\$100/month NAD stack	Walking + structured aerobic training	Better human-outcome evidence; also improves function
Repeated expensive “biological age” testing	BP + waist + fitness + strength + lipids + glucose	More actionable today
“Mitochondrial detox” product	Aerobic + resistance training	Direct, broad physiological adaptation
Large antioxidant stack	Fruit/vegetables + nuts + whole foods	Better nutritional value and evidence base
Longevity retreat	Sustainable weekly routine	Adherence over decades beats short bursts of optimization

Minimal Singapore protocol

Move daily • 2× strength/week • 150+ min aerobic/week • mostly Quarter–Quarter–Half meals • less gravy/salty soup • healthy waist + muscle • 7–9 h sleep • know BP/lipids/glucose • no smoking • minimize alcohol/UV • screen/vaccinate when due • move workouts indoors when heat/haze is poor.

SECTION 17

Evidence notes + references

A short source set supporting the framework. Guidelines and systematic reviews were prioritized over individual mechanistic studies.

Interpretation notes

- Associations such as step count and mortality can be strong and useful but remain observational; effect sizes should not be presented as guaranteed causal benefit for an individual.
- Biomarkers and pathways are supporting evidence. They are not automatically validated surrogate endpoints for lifespan or disease prevention.
- Medication targets, screening intervals, nutrient doses and disease-specific recommendations must be individualized by clinicians when relevant.
- Singapore prices, subsidies and public-health rules can change; local program details are dated to the latest source reviewed in August 2026.

- [1] López-Otín C, et al. Hallmarks of Aging: An Expanding Universe. *Cell*. 2023;186:243–278. [Link](#)
- [2] World Health Organization. Guidelines / fact sheet on physical activity: adults 150–300 min moderate activity and muscle strengthening ≥2 days/week. [Link](#)
- [3] Ding D, et al. Daily steps and health outcomes in adults: systematic review and dose-response meta-analysis. *Lancet Public Health*. 2025;10:e668–e681. DOI 10.1016/S2468-2667(25)00164-1. [Link](#)
- [4] Momma H, et al. Muscle-strengthening activities and lower risk/mortality in major NCDs: systematic review and meta-analysis. *Br J Sports Med*. 2022;56:755–763. [Link](#)
- [5] Ettehad D, et al. Blood pressure lowering for prevention of cardiovascular disease and death: systematic review and meta-analysis. *Lancet*. 2016;387:957–967. [Link](#)
- [6] Ference BA, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease: EAS consensus statement. *Eur Heart J*. 2017;38:2459–2472. [Link](#)
- [7] World Cancer Research Fund. Cancer Prevention Recommendations. [Link](#)
- [8] World Health Organization. Sodium reduction fact sheet (updated 11 May 2026): adults <2,000 mg sodium/day. [Link](#)
- [9] World Health Organization. Healthy diet guidance: prioritize whole grains, vegetables, fruit and pulses. [Link](#)
- [10] American Heart Association. Life's Essential 8 – Healthy Sleep: most adults 7–9 hours/night. [Link](#)
- [11] Morton RW, et al. Protein supplementation and resistance-training gains: systematic review/meta-regression. *Br J Sports Med*. 2018;52:376–384. [Link](#)
- [12] Desai I, et al. Creatine supplementation and resistance-training body-composition changes: systematic review/meta-analysis. *J Strength Cond Res*. 2024;38:1813–1821. [Link](#)
- [13] Kraus WE, et al. Two years of calorie restriction and cardiometabolic risk (CALERIE). *Lancet Diabetes Endocrinol*. 2019. [Link](#)
- [14] Waziry R, et al. Caloric restriction and DNA-methylation measures of biological aging in CALERIE. *Nature Aging*. 2023;3:248–257. [Link](#)
- [15] Lee DJW, Hodzic Kuerec A, Maier AB. Targeting ageing with rapamycin and derivatives in humans: systematic review. *Lancet Healthy Longev*. 2024;5:e152–e162. [Link](#)
- [16] Life Biosciences. First participant dosed in Phase 1 trial of ER-100 partial epigenetic reprogramming for optic neuropathies, 9 Jun 2026. Company announcement; not peer-reviewed efficacy evidence. [Link](#)
- [17] Sehgal R, et al. Responsiveness of epigenetic aging biomarkers to longevity interventions in humans. *Nature Medicine*. Published 21 Aug 2026. DOI 10.1038/s41591-026-04562-9. [Link](#)
- [18] Singapore MOH/HPB. National Population Health Survey 2024 (updated chapters Feb 2026). [Link](#)
- [19] Singapore Health Promotion Board. National Nutrition Survey 2022. [Link](#)
- [20] HealthHub / HPB. My Healthy Plate – Quarter, Quarter, Half. [Link](#)
- [21] HealthHub. Eat Less / Healthy Hawker guidance: less sauce/gravy; avoid finishing salty soup. [Link](#)

- [22] HealthHub. Healthier SG Screening — subsidized screening fees and coverage. [Link](#)
- [23] Meteorological Service Singapore. Manage Heat Stress When Outdoors (2026 WBGT guidance). [Link](#)
- [24] National Environment Agency. Ultraviolet Index and sun-protection guidance. [Link](#)
- [25] National Environment Agency. Managing Haze / PSI and 1-hour PM2.5 guidance. [Link](#)
- [26] Singapore Ministry of Health. Principal Causes of Death — 2024. [Link](#)
- [27] ActiveSG. Individual Rates — ActiveGYM rates; last updated 6 Aug 2026. [Link](#)
- [28] ACC/AHA. 2026 Guideline on the Management of Dyslipidemia — Lp(a) at least once, selective ApoB. [Link](#)
- [29] Furbatto M, et al. Mediterranean Diet in Older Adults: systematic review/meta-analysis. *Nutrients*. 2024;16:3947. [Link](#)
- [30] Singapore HealthHub. Healthy Hawker Eats. [Link](#)

Closing synthesis

The operating principle

Start with what saves lives. Build functional reserve. Understand aging biology. Optimize only after the basics are strong. Explore frontier science last — and upgrade it only when credible human evidence arrives.